

IF3170 Inteligensi Artifisial

Institut Teknologi Bandung

Support Vector Machines

Semester I TA 2026/2027

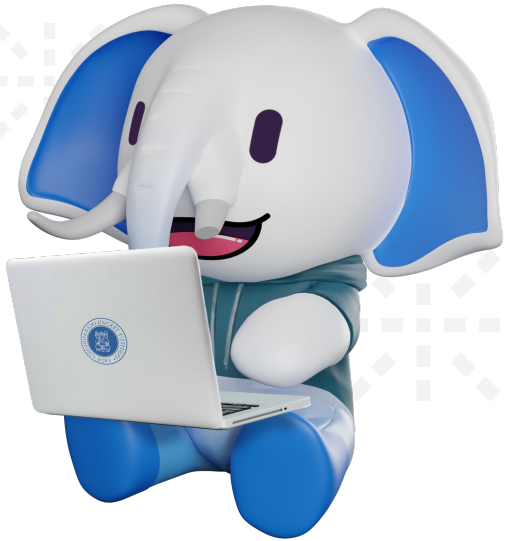


Outline

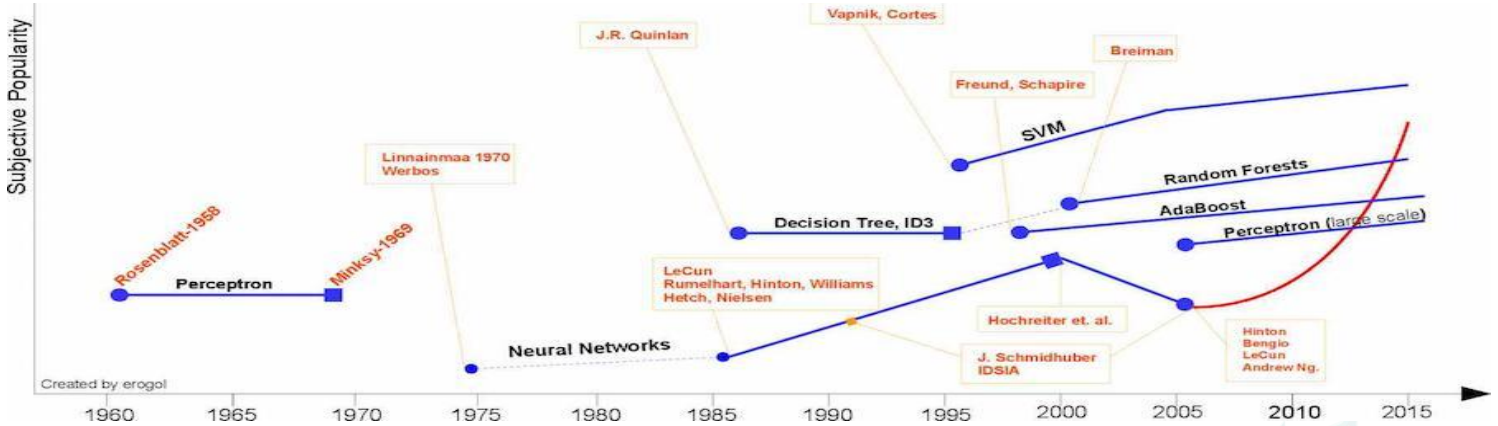
- Review:
    - Learning Agent
    - Introduction to Data Science
    - CRISP-DM
    - Data Understanding
    - Data Preparation
    - Decision Tree Learning
    - K-Nearest Neighbor
    - Linear & Logistic Regression

- Support Vector Machines

SVM: What & Why?



Support Vector Machine



- SVM diperkenalkan tahun 1992 oleh Vapnik, Boser, & Guyon
- Kinerja baik di berbagai aplikasi seperti bioinformatics, klasifikasi teks, pengenalan tulisan tangan dan lain-lain.

SVM Introduction

- 1980an
  - DTL dan NN memungkinkan pembelajaran nonlinear yang efisien
  - Kurang didukung dasar teoritis dan memungkinkan terjadinya local minima
- 1990an
  - Algoritma pembelajaran yang efisien untuk fungsi non linier berbasis teori komputasi

- Boser, B. E., Guyon, I. M., & Vapnik, V. N. (1992, July). A training algorithm for optimal margin classifiers. In *Proceedings of the fifth annual workshop on Computational learning theory* (pp. 144-152). ACM.
- Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine learning*, *20*(3), 273-297.

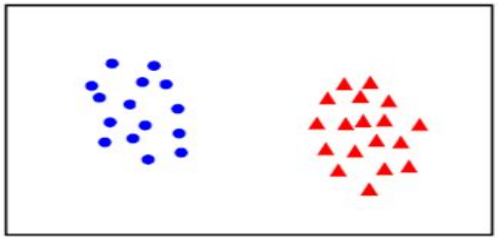
Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

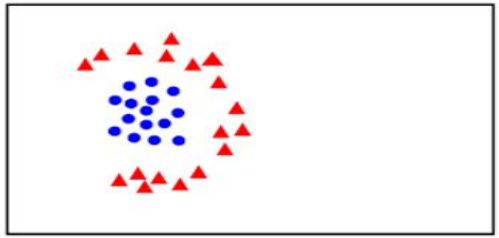
Klasifikasi Biner

Given training data (x_i, y_i) for $i = 1, \ldots, N$, with $x_i \in \mathbb{R}^d$ and $y_i \in \{-1, 1\}$, learn a classifier $f(x)$such that

$$f(x_i) \begin{cases} \geq 0, & y_i = +1 \\ < 0, & y_i = -1 \end{cases}$$

i.e., $y_i \, f(x_i) > 0$ for a correct classification.

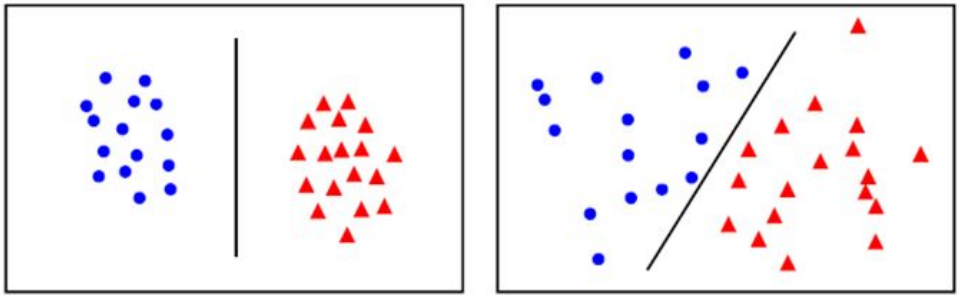




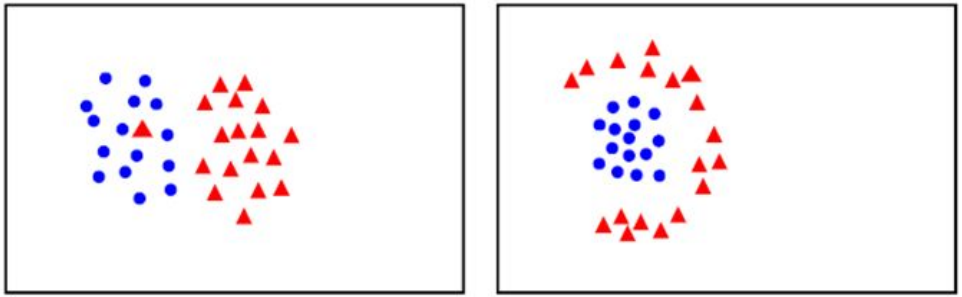
Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Linear Separability

linearly separable



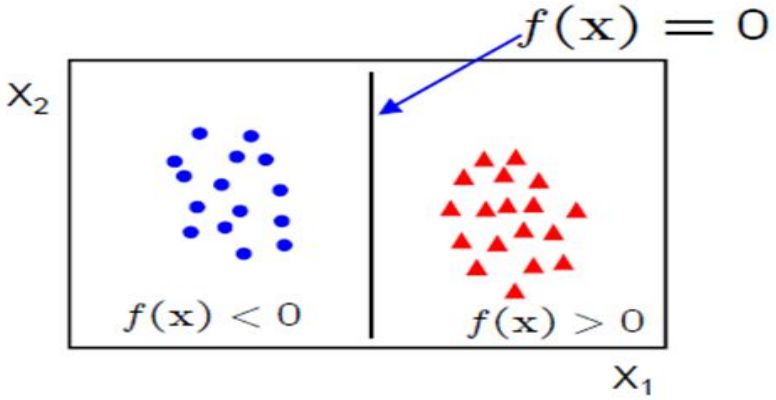
not linearly separable



Linear Classifier

- A linear classifier has the form

$$f(x) = w^\top x + b$$



- In 2D the discriminant is a line
- **w** is the normal to the line, and b the bias
- **w** is known as the weight vector

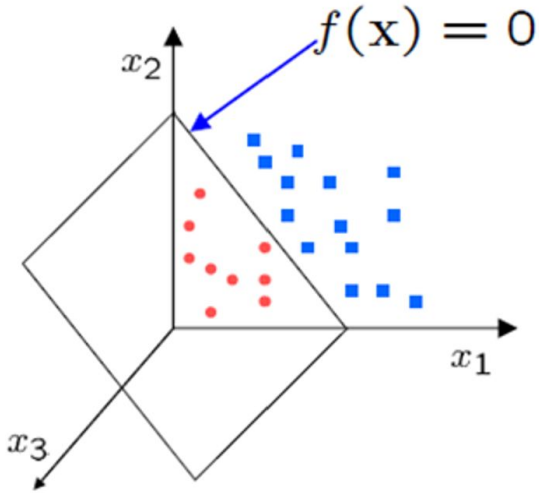
Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Linear Classifier

- A linear classifier has the form

$$f(x) = w^\top x + b$$

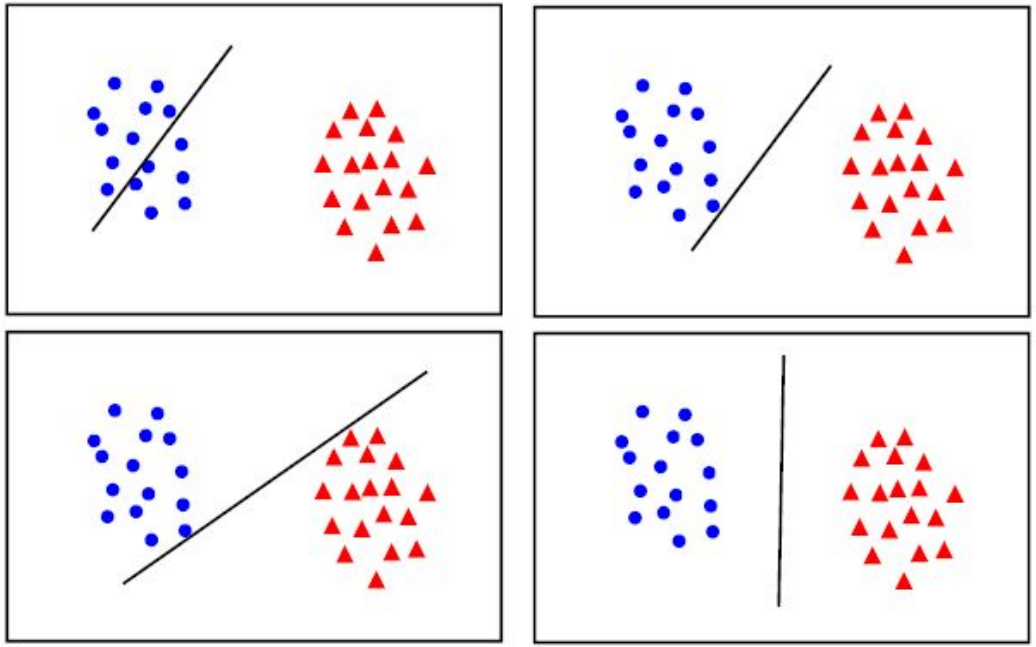
- in 3D the discriminant is a plane, and in nD it is a hyperplane



- For a K-NN Classifier it was necessary to 'carry' the training data
- For a linear classifier, the training data is used to learn **w** and then discarded
- Only **w** is needed for classifying new data.

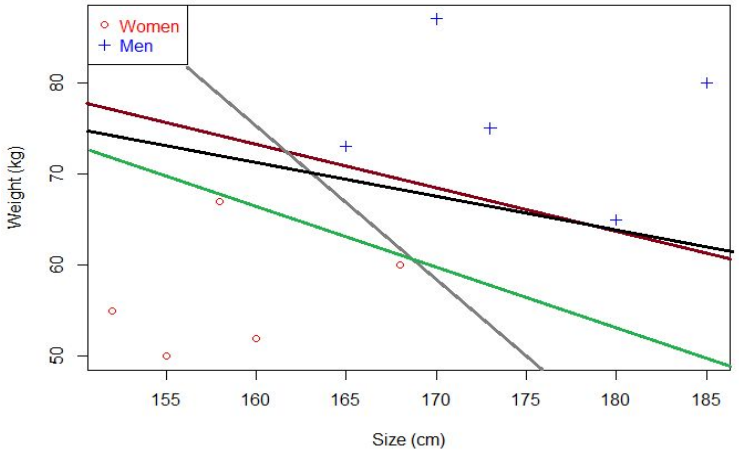
Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

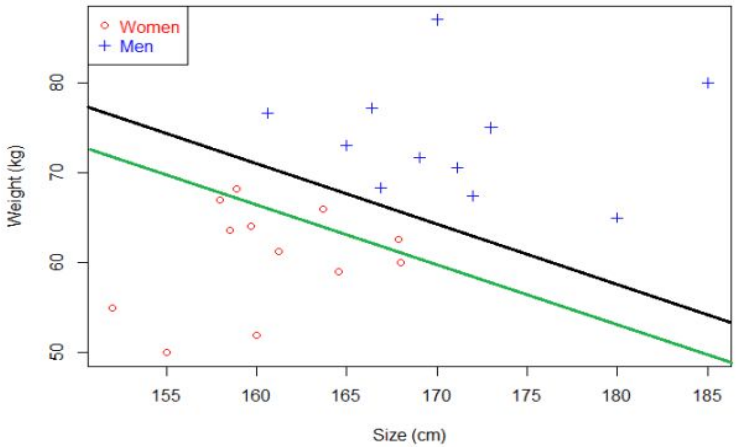
Bidang Pemisah Terbaik?



Mengapa?

Bidang Pemisah Terbaik (lanj)





- Kiri: semua bidang pemisah valid karena memisahkan kedua kelas pada training data.
- Kanan: real-life data. Bidang pemisah hitam lebih baik daripada hijau.

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

SVM Objective

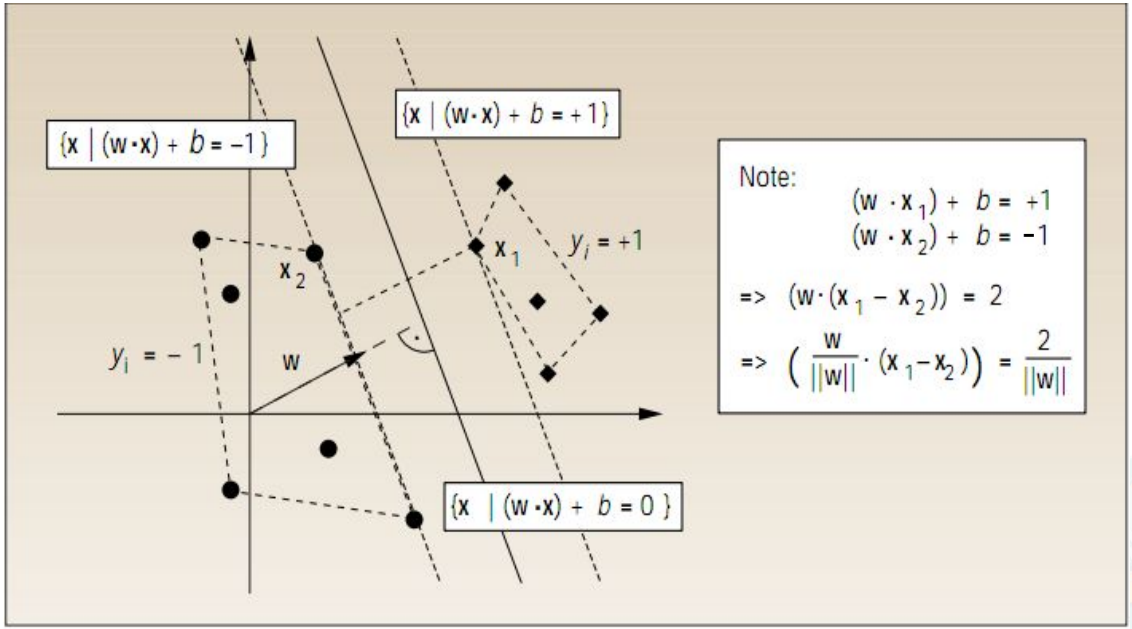
- Objective of the SVM **is to find the optimal separating hyperplane which maximizes the margin of the training data.** There will never be any data point inside the margin.
- Menggunakan optimasi kuadratik untuk menghindari 'local minimum' isu yang ada pada NN (Greedy)
- Menggunakan fungsi kernel untuk memisahkan non-linear region

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Hyperplane Classifier

- Hipotesis:
- x1, x2 ∈ training data

$$f(x) = sign(\vec{w}.\vec{x} + b); \vec{w}, \vec{x} \in \Re^N; b \in \Re$$



Revisit: Vector Direction

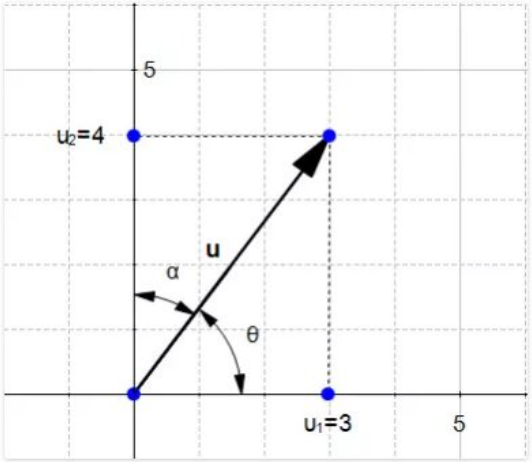


Figure 4 - direction of a vector

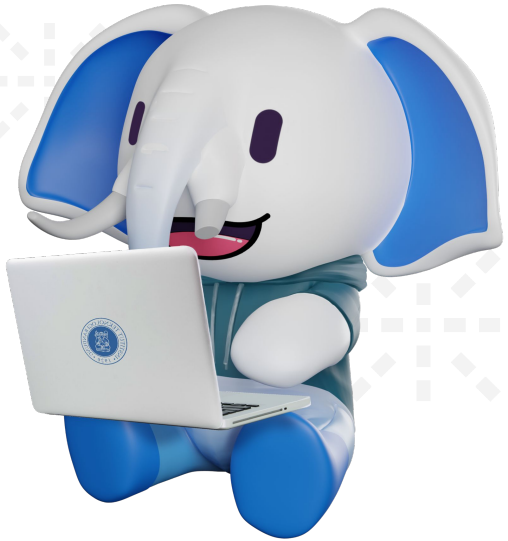
$\boldsymbol{u}(u_1, u_2)$ with $u_1 = 3$ and $u_2 = 4$

$$\cos(\theta) = \frac{u_1}{\|u\|} \qquad \cos(\alpha) = \frac{u_2}{\|u\|}$$

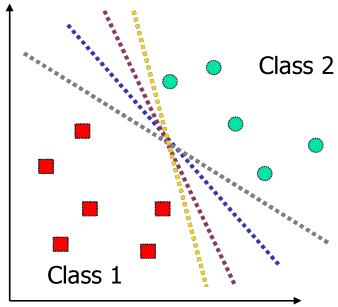
- Naive definition 1: The direction of the vector u is defined by the angle θ with respect to the horizontal axis, and with the angle α with respect to the vertical axis.
- Naive definition 2: The direction of the vector u is defined by the cosine of the angle θ and the cosine of the angle α.

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

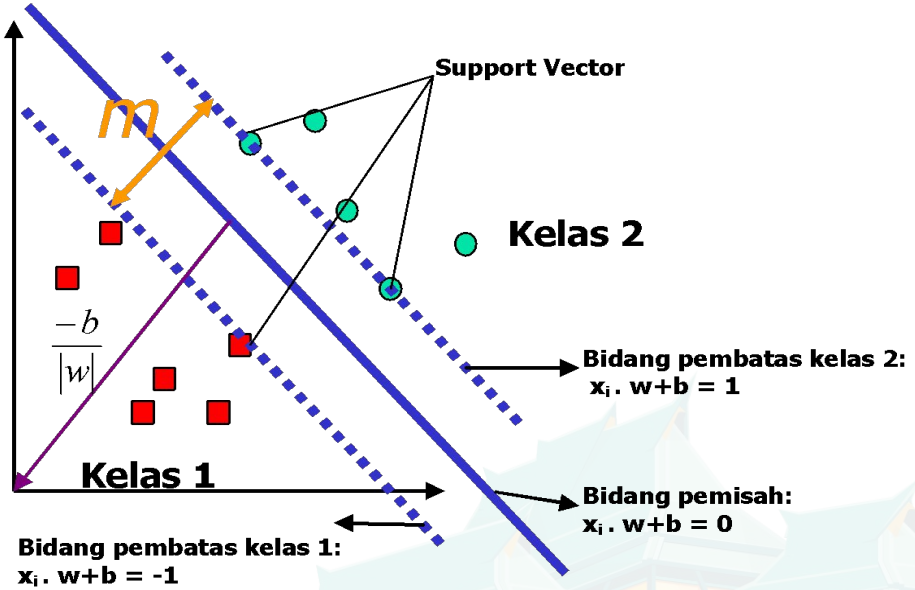
SVM for Linearly Separable Data



Mencari Bidang Pemisah Terbaik (1)

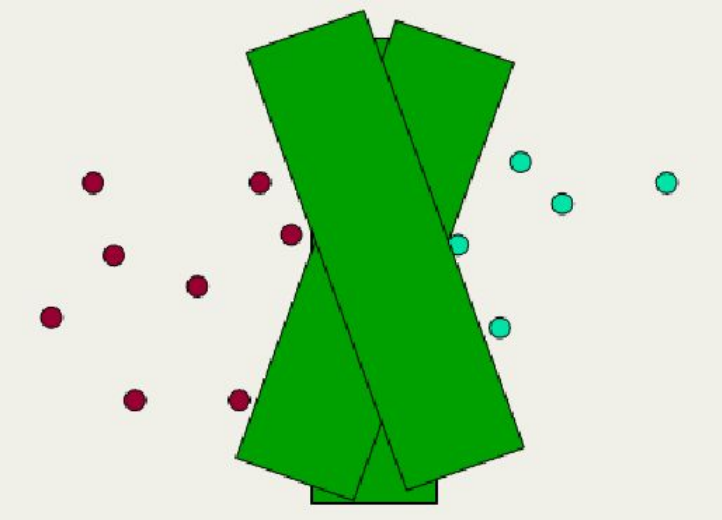


Mencari bidang pemisah dengan margin (m) terbesar



Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Mencari Bidang Pemisah Terbaik (2)



- 2 kelas dapat dipisahkan oleh sepasang bidang pembatas yang sejajar.
  - Bidang pembatas pertama membatasi kelas pertama
  - Bidang pembatas kedua membatasi kelas kedua
- Support Vector : Vector pada training data yang men-support bidang pemisah

Mengapa mencari margin terbesar?

- Bidang pemisah terbaik dengan margin terbesar memiliki generalisasi yang lebih baik
  - Titik yang dekat dengan bidang pemisah merepresentasikan ketidakyakinan klasifikasi : 50% peluang pengambilan keputusan oleh classifier
- Kapasitas memori untuk menyimpan model menjadi lebih sedikit
  - Hanya support vector yang mempengaruhi pengambilan keputusan

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

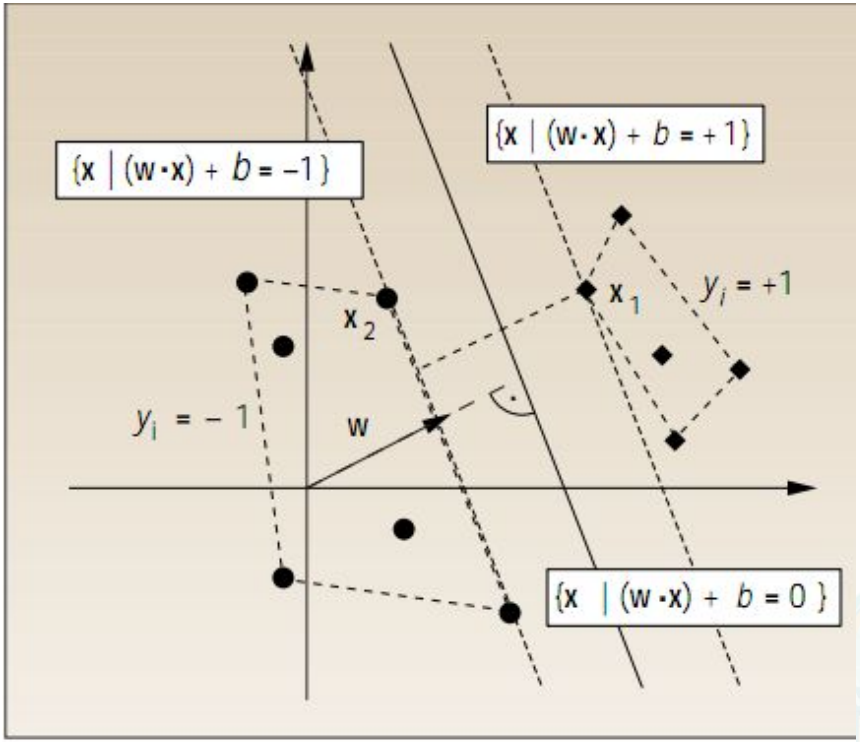
Support Vectors

- Support vectors memenuhi:

$$| \vec{w}.\vec{x}_i + b | = 1$$

- Untuk itu, semua training data berlabel "+" dan "–" memenuhi juga:

$$y_i(\vec{w}.\vec{x}_i + b) \geq 1$$



Optimal Hyperplane

- Optimal hyperplane:
  - maximum-margin hyperplane
  - atau hyperplane with maximal margin separation between two classes.
- Maximal margin (2/||w||) ≈ minimize ||w|| dengan batasan konsistensi training data tetap terjaga (semua klasifikasi benar).
- Minimize:

$$V(\vec{w}, b) = \frac{1}{2}\vec{w}.\vec{w}$$

subject to:

$$y_i(\vec{w}.\vec{x}_i + b) \geq 1$$

Recall the distance from a point (x_0, y_0) to a line: $Ax + By + c = 0$ is

$$\frac{|Ax_0 + By_0 + c|}{\sqrt{A^2 + B^2}}$$

The distance between H and H_1 is:

$$\frac{|w \cdot x + b|}{\| w \|} = \frac{1}{\| w \|}$$

The distance between H_1 and H_2 is:

$$\frac{2}{\| w \|}$$

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Optimal Hyperplane

- Pencarian bidang pemisah terbaik dapat dirumuskan menjadi:

$$\min_{w,\,b} \ \frac{1}{2} \parallel w \parallel^2$$

s.t. $y_i\left(x_i^\top w + b\right) - 1 \geq 0$

- Supaya lebih mudah diselesaikan ubah ke formula Lagrangian

$$\min_{w,\,b} \ L_p(w,b,\alpha) = \frac{1}{2} \parallel w \parallel^2 - \sum_{i=1}^{n} \alpha_i \ y_i\left(x_i^\top w + b\right) + \sum_{i=1}^{n} \alpha_i$$

with $\alpha_i \geq 0$

- Minimumkan L_p terhadap b dan w (saddle point), diperoleh:

$$\frac{\partial}{\partial b} L_p(w,b,\alpha) = 0 \ \rightarrow \ \boxed{\sum_{i=1}^{n} \alpha_i \, y_i = 0}$$

dan

$$\frac{\partial}{\partial w} L_p(w,b,\alpha) = 0 \ \rightarrow \ \frac{\partial L_p}{\partial w} = w - \sum_{i=1}^{n} \alpha_i \, y_i x_i = 0 \ \rightarrow \ \boxed{w = \sum_{i=1}^{n} \alpha_i \, y_i x_i}$$

- Want to look for solution point p where

$$\nabla f(p) = \nabla \lambda g(p)$$

$$g(x) = 0$$

- Or, combining these two as the *Langrangian* L & requiring derivative of L be zero:

$$L(x,a) = f(x) - ag(x)$$

$$\nabla(x,a) = 0$$

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Revisit: Lagrangian Dual Problem

- Pada **masalah dual Lagrangian**, alih-alih melakukan minimisasi terhadap w dan b dengan kendala yang melibatkan nilai-nilai α, kita dapat melakukan maksimisasi terhadap α(variabel dual) dengan memenuhi relasi yang sebelumnya telah diperoleh untuk w dan b.
- Solusi yang diperoleh harus memenuhi dua relasi berikut:

$$w = \sum_{i=1}^{l} \alpha_i \, y_i x_i \quad \text{dan} \quad \sum_{i=1}^{l} \alpha_i \, y_i = 0$$

- Dengan mensubstitusikan kembali w dan b ke dalam persamaan awal, kita dapat menghilangkan ketergantungan terhadap w dan b.
- Perhatikan bahwa pada tahap ini kita sebenarnya sudah mengetahui bentuk solusi untuk bobot w, yaitu bahwa w merupakan kombinasi linear dari data masukan pelatihan (x_i), label pelatihan (y_i), dan nilai α_i.
- Selanjutnya kita mencari nilai α_i dengan mendiferensiasikan fungsi dual terhadap α_i, kemudian menyamakan hasil turunannya dengan nol.
- Sebagian besar nilai α_i akan bernilai **nol**.
- Nilai α_i yang **tidak nol** akan terkait dengan **support vector**.

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Substitute the Primal Problem

- Primal problem

$$\min_{w,b} L_P = \frac{1}{2} \| w \|^2 - \sum_{i=1}^{l} \alpha_i y_i (x_i^\top w + b) + \sum_{i=1}^{l} \alpha_i$$

s.t. $\forall i, \alpha_i \geq 0$

$$w = \sum_{i=1}^{l} \alpha_i y_i x_i, \qquad \sum_{i=1}^{l} \alpha_i y_i = 0$$

- Dual problem

$$\max_{\alpha} L_D(\alpha_i) = \sum_{i=1}^{l} \alpha_i - \frac{1}{2} \sum_{j=1}^{l} \alpha_i \alpha_j y_i y_j (x_i^\top x_j)$$

s.t. $\sum_{i=1}^{l} \alpha_i y_i = 0$ dan $\alpha_i \geq 0$

(note that we have removed the dependence on wand b)

Quadratic Optimization Problem

- **Minimize:** $W(\vec{\alpha}) = -\sum_{i=1}^{n} \alpha_i + \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} y_i \, y_j \, \alpha_i \alpha_j (\overrightarrow{x_i} . \overrightarrow{x_j})$

  subject to: $\sum_{i=1}^{n} y_i \, \alpha_i = 0, \forall i \in \{1, \ldots, n\} : \alpha_i \geq 0$

- **Support vectors:** $\alpha_i > 0$

- **Hyperplane:** $\vec{w} . \vec{x} = (\sum_{i=1}^{n} \alpha_i \, y_i \overrightarrow{x_i}) . \vec{x} = \sum_{i=1}^{n} \alpha_i \, y_i (\overrightarrow{x_i} . \vec{x})$

$$b = y_{sv} - \vec{w} . \overrightarrow{x_{sv}}$$

Optimization Problem Example

Suppose we have two 2D data points: $[(x_{11}, x_{12}),\ (x_{21}, x_{22})]$ with labels $[y_1, y_2]$

- **Minimize:** $W(a) = -(a_1 + a_2) +$

$$\frac{1}{2}[y_1 y_1 a_1 a_1 (x_{11}x_{11} + x_{12}x_{12}) + y_1 y_2 a_1 a_2 (x_{11}x_{21} + x_{12}x_{22}) + y_2 y_1 a_2 a_1 (x_{21}x_{11} + x_{22}x_{12}) + y_2 y_2 a_2 a_2 (x_{21}x_{21} + x_{22}x_{22})]$$

$$W(\vec{\alpha}) = -\sum_{i=1}^{n} \alpha_i + \frac{1}{2}\sum_{i=1}^{n} \sum_{j=1}^{n} y_i\, y_j \alpha_i \alpha_j (\overrightarrow{x_i}.\overrightarrow{x_j})$$

subject to: $(y_1 a_1 + y_2 a_2) = 0$

$$\sum_{i=1}^{n} y_i\, \alpha_i = 0, \forall i \in \{1, \dots, n\}$$

- **Support vectors:** $a_1 > 0, a_2 > 0$

$$\alpha_i > 0$$

- **Hyperplane:**

$$w \cdot x = a_1 y_1 ((x_{11}, x_{12}) \cdot x) + a_2 y_2 ((x_{21}, x_{22}) \cdot x)$$

$$\vec{w}.\vec{x} = (\sum_{i=1}^{n} \alpha_i\, y_i \overrightarrow{x_i}).\vec{x} = \sum_{i=1}^{n} \alpha_i\, y_i (\overrightarrow{x_i}.\vec{x})$$

$$b = y_{sv} - w \cdot x_{sv}$$

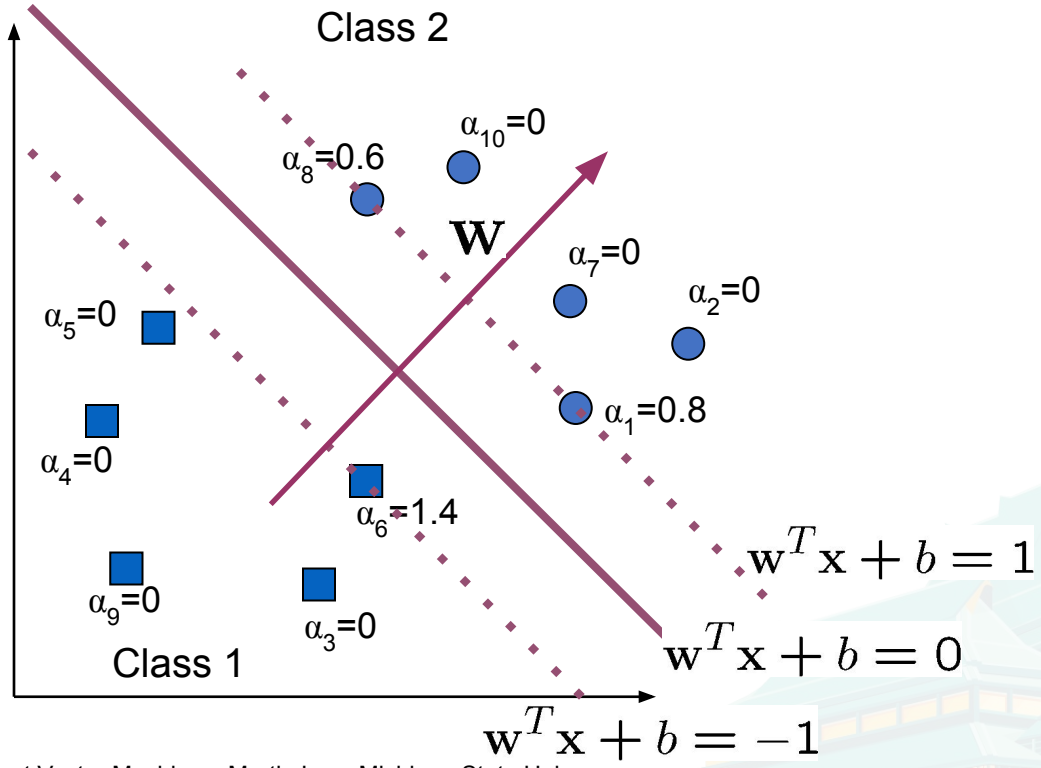
$$b = y_{sv} - \vec{w}.\overrightarrow{x_{sv}}$$

Revisit: Quadratic Programming

- Quadratic programming (QP) is the problem of optimizing a quadratic objective function and is one of the simplest form of non-linear programming.
- The objective function can contain bilinear or up to second order polynomial terms, and the constraints are linear and can be both equalities and inequalities.

$$\min_x f(x) = q^\top x + \frac{1}{2} x^\top Q x$$
$$s.t.\ Ax = a,$$
$$Bx \leq b,$$
$$x \geq 0.$$

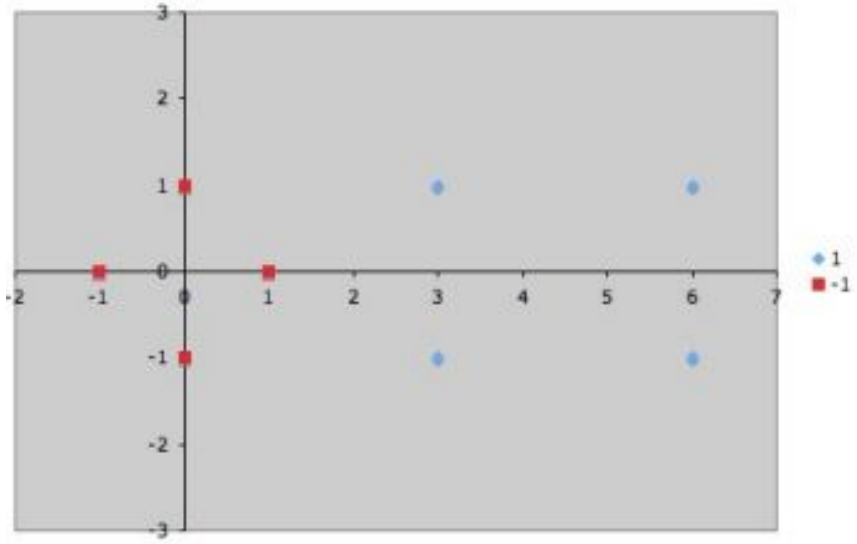
A Geometric Interpretation



A Simple Introduction to Support Vector Machines; Martin Law; Michigan State Univ.

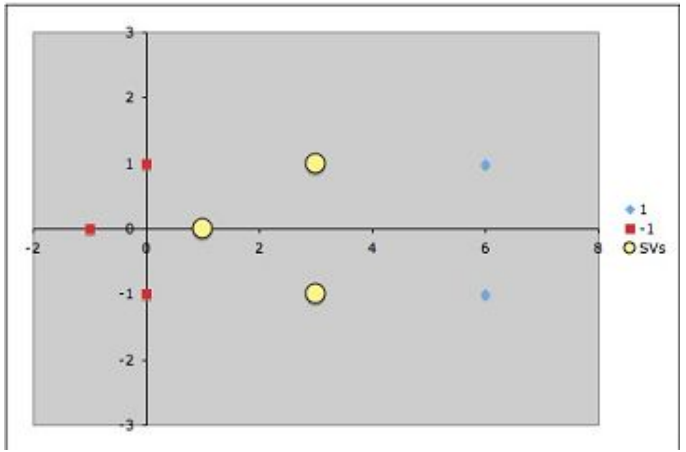
Contoh

x1	x2	Kelas
3	1	+1
3	-1	+1
6	1	+1
6	-1	+1
1	0	-1
0	1	-1
0	-1	-1
-1	0	-1



- Taken from: http://axon.cs.byu.edu/Dan/678/miscellaneous/SVM.example.pdf

Support Vectors



$$f(\vec{x}) = \sum_{i=1}^{nsv} (\alpha_i y_i \vec{x}_i \vec{x}) + b$$

$$\begin{pmatrix}1\\0\end{pmatrix} \Rightarrow -1 = \alpha_1.-1.\begin{pmatrix}1\\0\end{pmatrix}.\begin{pmatrix}1\\0\end{pmatrix} + \alpha_2.1.\begin{pmatrix}3\\1\end{pmatrix}.\begin{pmatrix}1\\0\end{pmatrix} + \alpha_3.1.\begin{pmatrix}3\\-1\end{pmatrix}.\begin{pmatrix}1\\0\end{pmatrix} + b = -\alpha_1 + 3\alpha_2 + 3\alpha_3 + b \ldots(1)$$

$$\begin{pmatrix}3\\1\end{pmatrix} \Rightarrow 1 = \alpha_1.-1.\begin{pmatrix}1\\0\end{pmatrix}.\begin{pmatrix}3\\1\end{pmatrix} + \alpha_2.1.\begin{pmatrix}3\\1\end{pmatrix}.\begin{pmatrix}3\\1\end{pmatrix} + \alpha_3.1.\begin{pmatrix}3\\-1\end{pmatrix}.\begin{pmatrix}3\\1\end{pmatrix} + b = -3\alpha_1 + 10\alpha_2 + 8\alpha_3 + b \ldots(2)$$

$$\begin{pmatrix}3\\-1\end{pmatrix} \Rightarrow 1 = \alpha_1.-1.\begin{pmatrix}1\\0\end{pmatrix}.\begin{pmatrix}3\\-1\end{pmatrix} + \alpha_2.1.\begin{pmatrix}3\\1\end{pmatrix}.\begin{pmatrix}3\\-1\end{pmatrix} + \alpha_3.1.\begin{pmatrix}3\\-1\end{pmatrix}.\begin{pmatrix}3\\-1\end{pmatrix} + b = -3\alpha_1 + 8\alpha_2 + 10\alpha_3 + b \ldots(3)$$

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

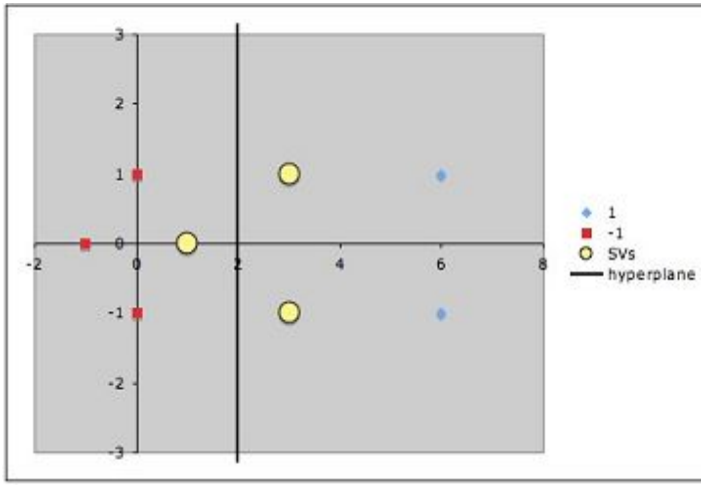
$$-\alpha_1 + 3\alpha_2 + 3\alpha_3 + b = -1 \ldots . (1)$$
$$-3\alpha_1 + 10\alpha_2 + 8\alpha_3 + b = 1 \ldots . (2)$$
$$-3\alpha_1 + 8\alpha_2 + 10\alpha_3 + b = 1 \ldots . (3)$$
$$-\alpha_1 + \alpha_2 + \alpha_3 = 0 \ldots . (4)$$

$$(2) - (3): 2\alpha_2 - 2\alpha_3 = 0 \rightarrow \alpha_2 = \alpha_3 \ldots (5)$$
$$(5) subs (1): -\alpha_1 + 6\alpha_2 + b = -1 \ldots . (6)$$
$$(5) subs (2): -3\alpha_1 + 18\alpha_2 + b = 1 \ldots . (7)$$
$$(6) - (7): 2\alpha_1 - 12\alpha_2 = -2$$
$$\alpha_1 - 6\alpha_2 = -1$$
$$\alpha_1 = 6\alpha_2 - 1 \ldots . (8)$$
$$(8,5) subs (4): -6\alpha_2 + 1 + \alpha_2 + \alpha_2 = 0$$
$$-4\alpha_2 = -1 \rightarrow \alpha_2 = \frac{1}{4} = 0.25$$

$$\alpha_1 = 0.5; \alpha_2 = 0.25; \alpha_3 = 0.25$$
$$substitusi (6): -0.5 + 6 * 0.25 + b = -1$$
$$b = -2$$

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Hipotesis

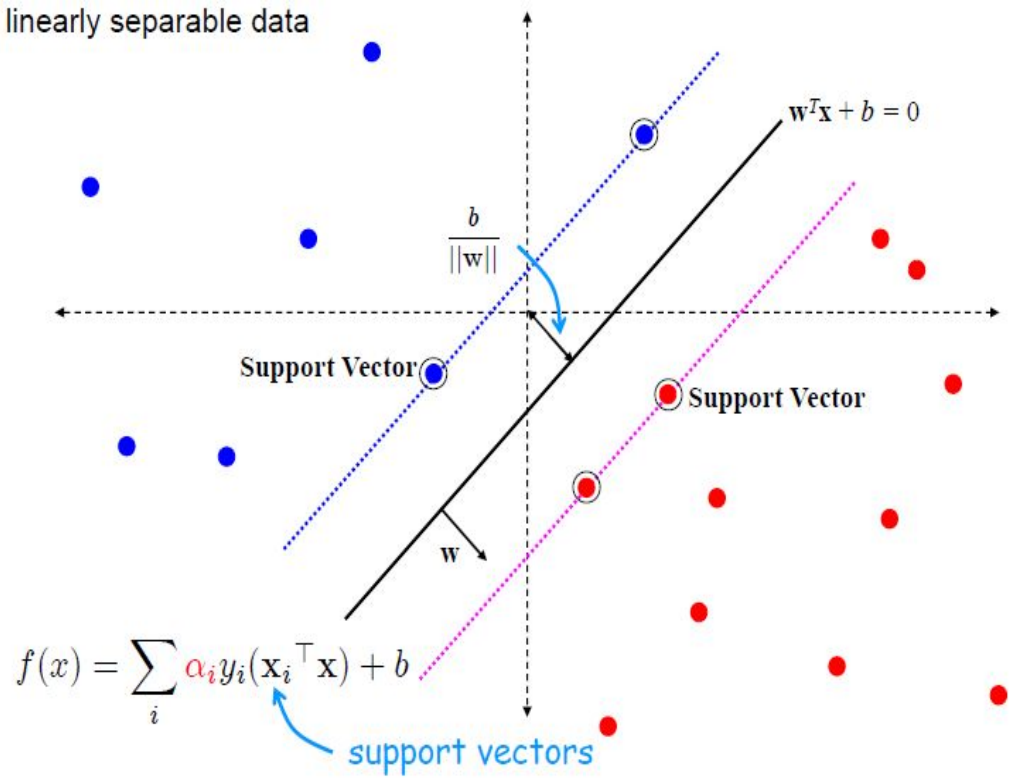


$$f(\vec{x}) = \sum_{i=1}^{nsv} (\alpha_i y_i \vec{x}_i \vec{x}) - 2; \alpha_1 = 0.5; \alpha_2 = \alpha_3 = 0.25$$

$$f\left(\begin{pmatrix}6\\1\end{pmatrix}\right) = sign((0.5)(-1)\begin{pmatrix}1\\0\end{pmatrix}.\begin{pmatrix}6\\1\end{pmatrix} + (0.25)(1)\begin{pmatrix}3\\1\end{pmatrix}.\begin{pmatrix}6\\1\end{pmatrix} + (0.25)(1)\begin{pmatrix}3\\-1\end{pmatrix}.\begin{pmatrix}6\\1\end{pmatrix} - 2) = sign(-3+4.75+4.25-2) = sign(4) = 1$$

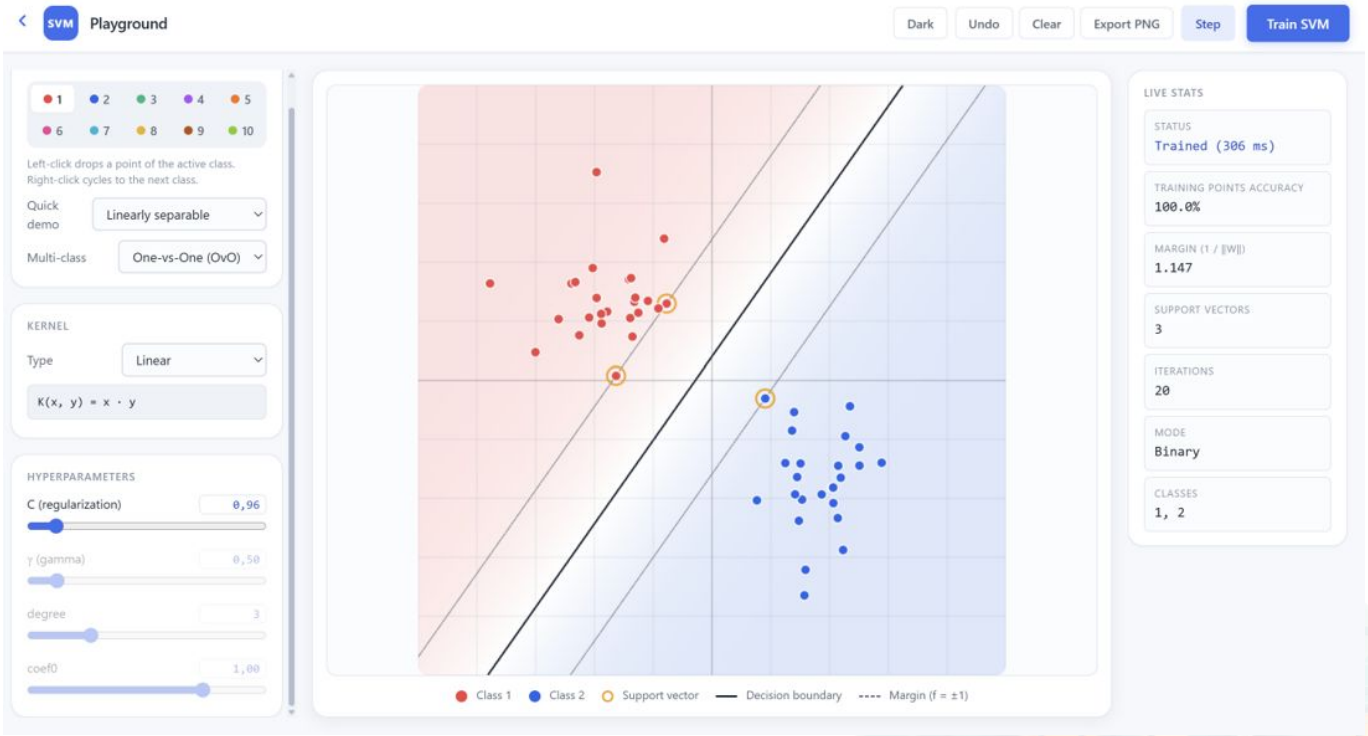
Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Summary



Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

SVM Simulation



https://svmplayground.vercel.app/simulation

Revisit: Inner Product and Similarity

Why should inner product kernels be involved in pattern recognition using SVMs, or at all?

- Intuition is that inner products provide some measure of *'similarity'*
- Inner product in 2D between 2 vectors of unit length returns the cosine of the angle between them = how *'far apart'* they are

  e.g. $x = [1, 0]^T, y = [0, 1]^T$

  i.e. if they are **parallel** their inner product is 1 (completely **similar**)

  $$x^T y = x \cdot y = 1$$

  If they are **perpendicular** (completely **unlike**) their inner product is 0 (so should not contribute to the correct classifier)

  $$x^T y = x \cdot y = 0$$

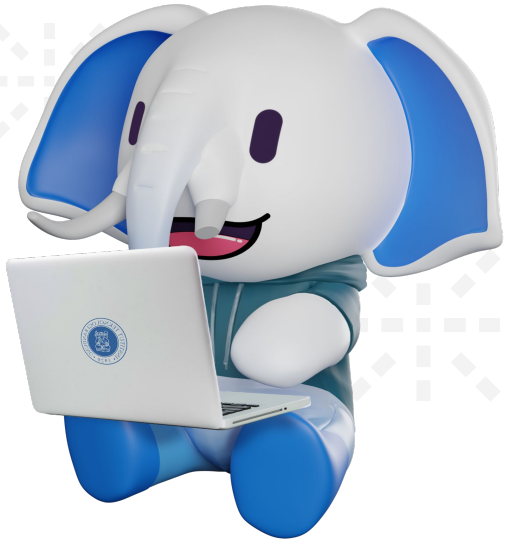
Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Latihan

1. Data: $x_1 = (3, 1)$ kelas +1 dan $x_2 = (1, 1)$ kelas −1. Tentukan α, w, b, persamaan bidang pemisah, dan margin. Klasifikasikan (2,5; 5).

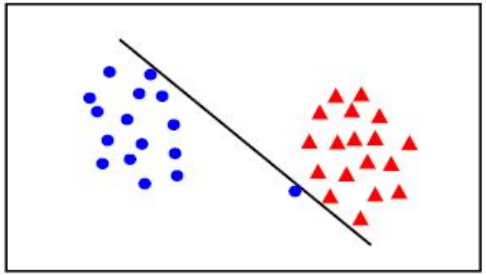
Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

SVM for Non-linearly Separable Data

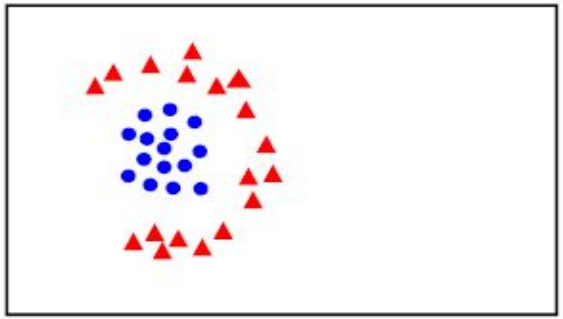


Non-Linearly Separable

- Existence of the noise



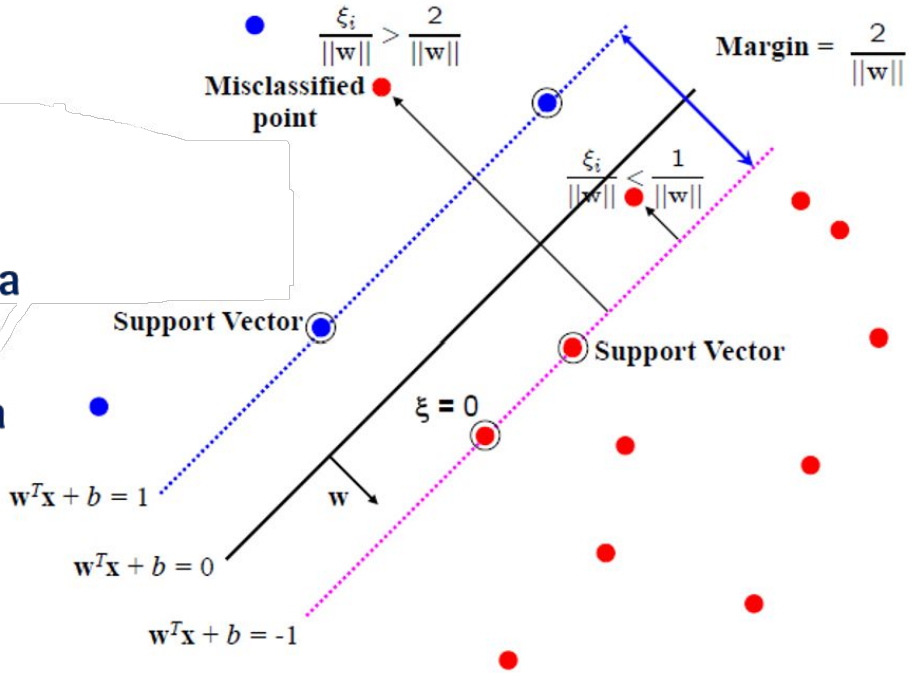
- Nature of data : Non-linear boundary



Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Slack Variable

- Slack variable: error dalam klasifikasi $\xi \geq 0$
- Untuk $0 < \xi \leq 1$ titik berada **di antara margin dan sisi yang benar dari hyperplane**. Kondisi ini disebut **margin violation.**
- Untuk $\xi > 1$ titik **salah klasifikasi (misclassified).**\



https://www.robots.ox.ac.uk/~az/lectures/ml/lect2.pdf

Noise in Data

Minimize: $\Phi(w) = \frac{1}{2} w^T w$

Subject to: $d_i(w^T x_i + b) \geq 1 \ \forall i$

↓

Introduce slack variables $\xi_i \geq 0$

↓

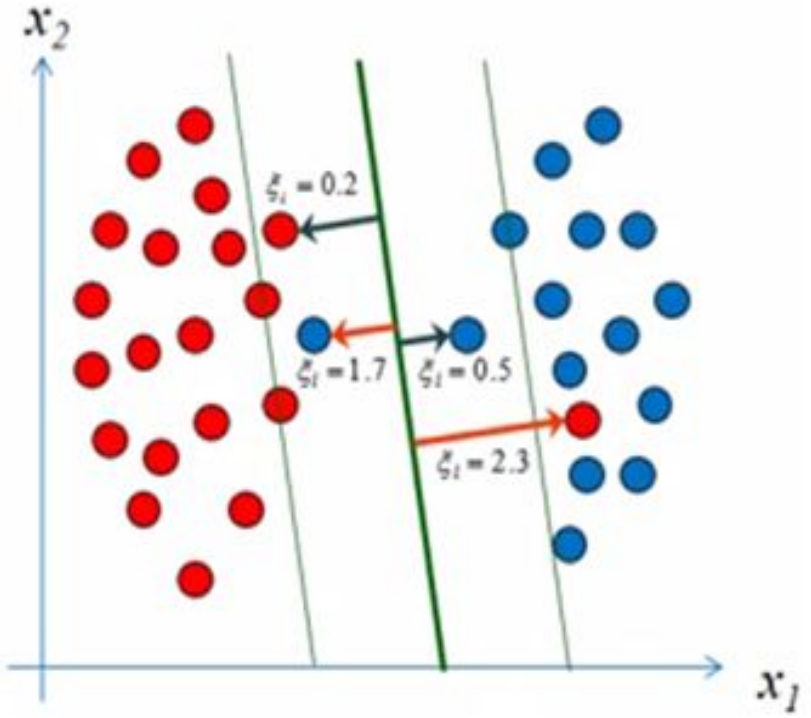
Minimize: $\Phi(w) = \frac{1}{2} w^T w$

Subject to: $d_i(w^T x_i + b) \geq 1 - \xi_i \ \forall i$

Also minimize training error $\sum_{i=1}^{N} I(\xi_i \geq 1)$ or $\sum_{i=1}^{N} \xi_i$

$$\Phi(w) = \frac{1}{2} w^T w + C \sum_{i=1}^{N} \xi_i$$

Subject to: $d_i(w^T x_i + b) \geq 1 - \xi_i; \ \xi_i \geq 0, \forall i$



Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

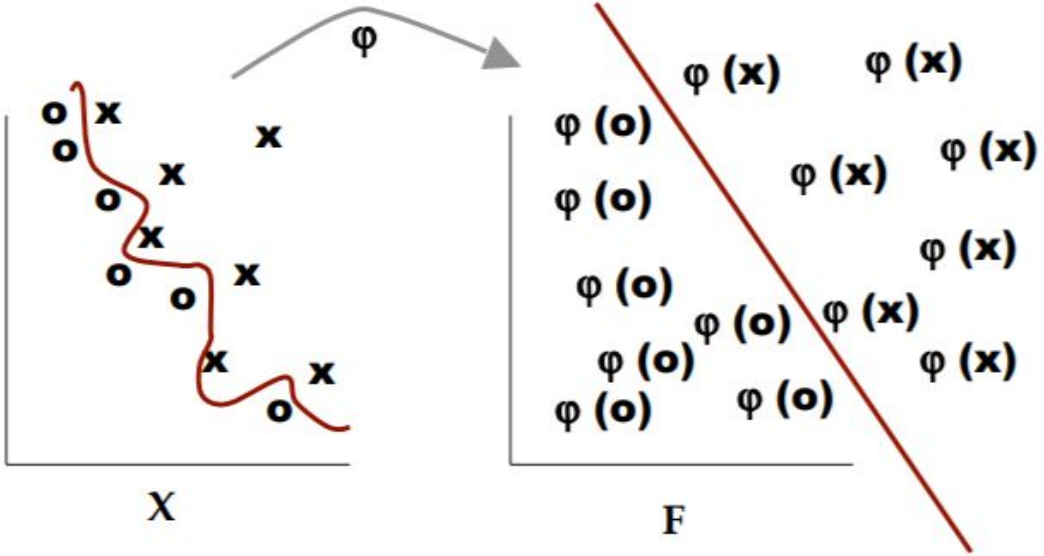
Dual Form dengan Slack

- **Forming the Lagrangian and Converting to the Dual**
  Dengan membentuk fungsi Lagrangian dan mengubahnya ke bentuk dual, diperoleh:
- $Q(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j d_i d_j x_i^T x_j$

  dengan kendala: $0 \leq \alpha_i \leq C, \forall i$ dan $\sum_{i=1}^{N} \alpha_i d_i = 0$

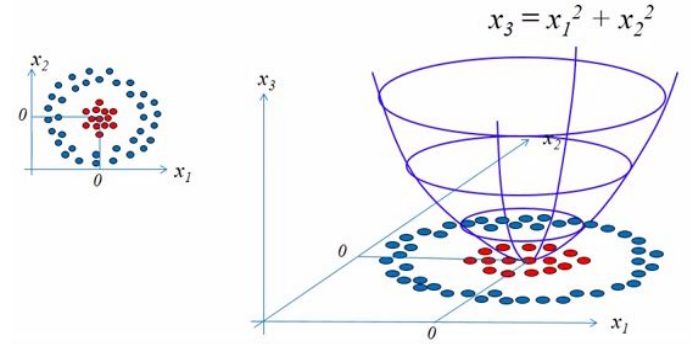
- **Catatan**
  - Perhatikan bahwa **slack variables** (ξ_i) maupun **Lagrange multiplier** yang terkait dengannya tidak muncul dalam formulasi dual.
  - Satu-satunya perubahan dibandingkan hard-margin SVM adalah adanya kendala tambahan pada α_i: $0 \leq \alpha_i \leq C$
    Parameter C mengontrol keseimbangan (*trade-off*) antara:
  - **Training error** (kesalahan pada data pelatihan)
  - **Model complexity / VC dimension** (kompleksitas model)

Non-Linear Boundary Transformation

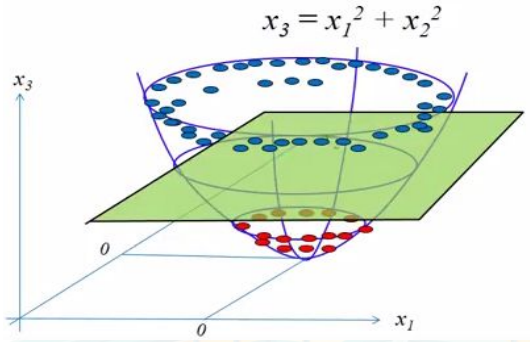


Non-Linear Boundary Transformation

- Lower Dimension: Non-linearly Separable Data

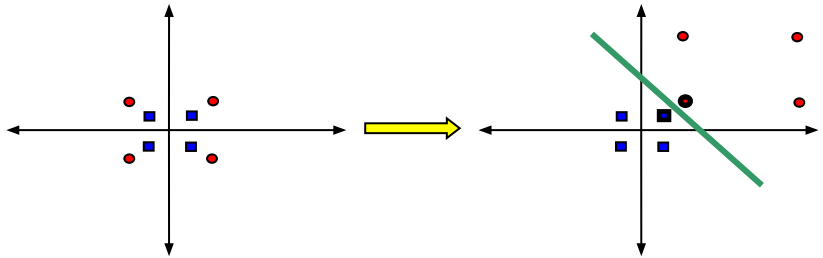


- Higher Dimension: Linearly Separable Data



SVM pada Non-linearly Separable Data

- Contoh
- Misalkan dataset:
  - Data kelas positif {(2,2),(2,-2),(-2,2),(-2,-2)}
  - Data kelas negatif {(1,1),(1,-1),(-1,1),(-1,-1)}



- Transformasi Fitur
- $$\phi(x_1, x_2) = \begin{cases} (4, -x_2 + \mid x_1 \mid - \mid x_2 \mid \;\; 4 - x_1 + \mid x_1 \mid - \mid x_2 \mid), & \sqrt{x_1^2 + x_2^2} > 2 \\ (x_1, x_2), & \sqrt{x_1^2 + x_2^2} \leq 2 \end{cases}$$

- Dengan transformasi diperoleh
  - Data kelas positif {(2,2),(6,2),(6,6),(2,6)}
  - Data kelas negatif {(1,1),(1,-1),(-1,1),(-1,-1)}

SVM pada Non-linearly Separable Data

- **Klasifikasi:**

$$f(x) = \sum_{i=1}^{ns} \alpha_i \, y_i x_i \cdot x + b \quad \Rightarrow \quad f(x) = \sum_{i=1}^{ns} \alpha_i \, y_i \phi(x_i)\phi(x) + b$$

- Sulit untuk mengetahui $\phi(x)$dan feature space biasanya memiliki dimensi yang lebih besar

- Solusinya *"kernel trick"*, yang perlu diketahui adalah $K(x_i, x) = \phi(x_i)\phi(x)$
- Dengan fungsi K (fungsi Kernel), maka fungsi $\phi(x)$ tidak perlu diketahui

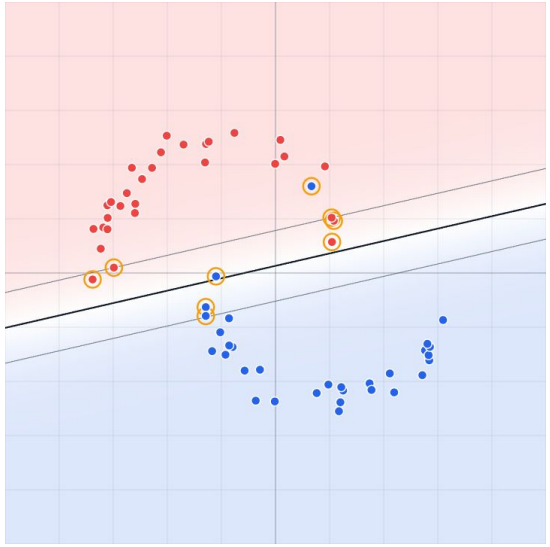
Klasifikasi:

$$f(x) = \sum_{i=1}^{ns} \alpha_i \, y_i K(x_i, x) + b$$

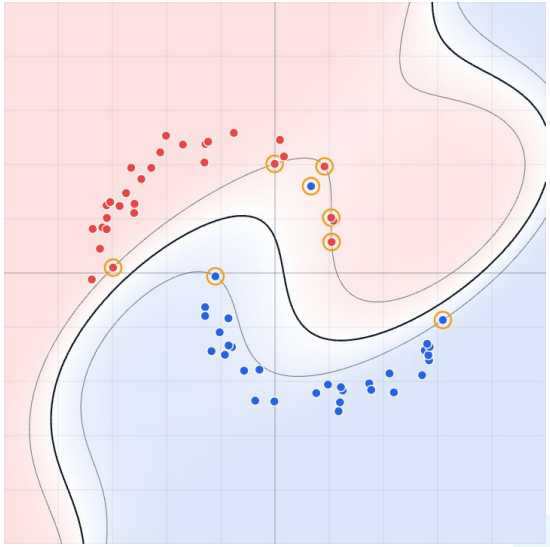
- Fungsi Kernel yang umum digunakan:

Linear Kernel	$K(x_i, x_j) = x_i^T x_j$
Polynomial kernel	$K(x_i, x_j) = (\gamma \cdot x_i^T x_j + r)^p, \gamma > 0$
RBF kernel	$K(x_i, x_j) = \exp(-\gamma \mid x_i - x_j \mid^2), \gamma > 0$
Sigmoid kernel	$K(x_i, x_j) = \tanh(\gamma x_i^T x_j + r)$

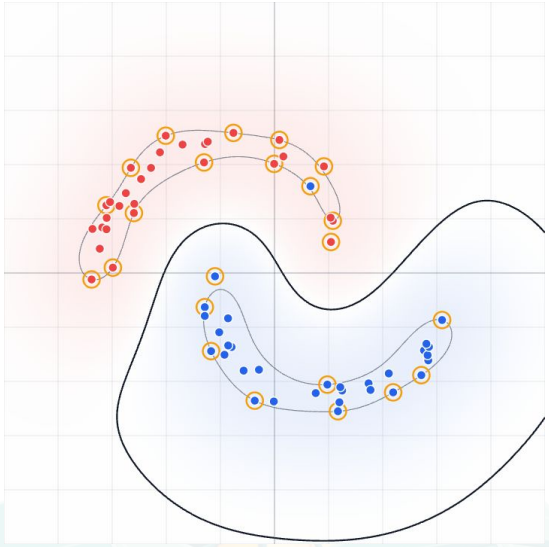
Different Kernel for Same Data



Linear



Polynomial



RBF

Example

- Suppose we have 5 (1D) data points
  - $x_1 = 1,\ x_2 = 2,\ x_3 = 4,\ x_4 = 5,\ x_5 = 6,$
  - with 1, 2, 5 as class 1 and 3, 4 as class 2 ⇒ $y_1 = 1,\ y_2 = 1,\ y_3 = -1,\ y_4 = -1,\ y_5 = 1$
- We use the polynomial kernel of degree 2
  - $K(x, y) = (xy + 1)^2$
  - C is set to 100
- We first find $a_i (i = 1, \dots, 5)$ by

$$\max. \sum_{i=1}^{5} \alpha_i - \frac{1}{2} \sum_{i=1}^{5} \sum_{j=1}^{5} \alpha_i \alpha_j y_i y_j (x_i x_j + 1)^2$$

$$\text{subject to } 100 \geq \alpha_i \geq 0, \sum_{i=1}^{5} \alpha_i y_i = 0$$

Example

- $x_1=1$, $x_2=2$, $x_3=4$, $x_4=5$, $x_5=6$,
- $y_1=1$, $y_2=1$, $y_3=-1$, $y_4=-1$, $y_5=1$

- By using a QP solver, we get
  - $\alpha_1 = 0,\ \alpha_2 = 2.5,\ \alpha_3 = 0,\ \alpha_4 = 7.333,\ \alpha_5 = 4.833$
  - Note that the constraints are indeed satisfied
  - The support vectors are $\{x_2, =, 2\ x_4 = 5\ x_5 = 6\}$
- The discriminant function is

α_5 y_5 $K(z, x_5)$

$$f(z) = 2.5(1)(2z+1)^2 + 7.333(-1)(5z+1)^2 + 4.833(1)(6z+1)^2 + b$$
$$= 0.6667z^2 - 5.333z + b$$

- b is recovered by solving $f(2) = 1$ or by $f(5) = -1$ or by $f(6) = 1$, as x_2 and x_5 lie on the line $\phi(w)^T\phi(x) + b = 1$ and x_4 lies on the line $\phi(w)^T\phi(x) + b = -1$

- All three give $b = 9$

  ➔ $f(z) = 0.6667z^2 - 5.333z + 9$

A Simple Introduction to Support Vector Machines; Martin Law; Michigan State Univ.

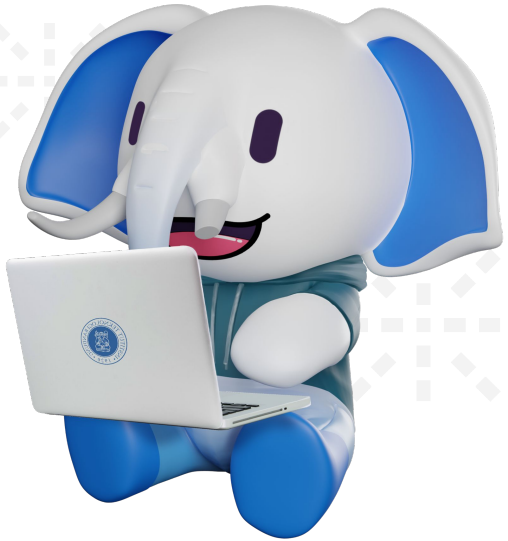
Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Latihan

1. Jelaskan efek menaikkan C dari 0.1 menjadi 1000 terhadap margin dan risiko overfitting.
2. Kelas A: $x = 1, 5$. Kelas B: $x = 2, 3, 4$. Cari $\varphi(x)$ agar data terpisah linear di ruang baru dan tentukan batasnya.

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

SVM for Multi-class Data



Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Multi Class SVM

- SVM dirancang untuk mengklasifikasikan data ke dalam dua kelas ⟶ SVM biner
- Untuk klasifikasi multiclass gabungkan beberapa SVM biner.
- Ada 3 metode umum:
  - *One-against-all*
  - *One-against-one*
  - DAG-SVM

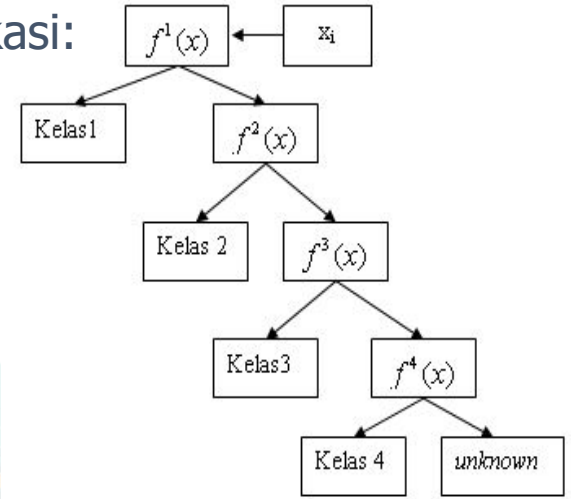
Multi Class SVM: One-against-all

- Dibangun k model klasifikasi (k adalah jumlah kelas)
- Setiap pelatihan model menggunakan data dari semua kelas
- Prediksi kelas data umumnya seperti gambar sebelah kanan atau berdasarkan nilai maksimum f(X)

Pembelajaran:

$y_i = 1$	$y_i = -1$	Hipotesis
Kelas 1	Bukan kelas 1	$f^1(x) = (w^1)x + b^1$
Kelas 2	Bukan kelas 2	$f^2(x) = (w^2)x + b^2$
Kelas 3	Bukan kelas 3	$f^3(x) = (w^3)x + b^3$
Kelas 4	Bukan kelas 4	$f^4(x) = (w^4)x + b^4$

Klasifikasi:



Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

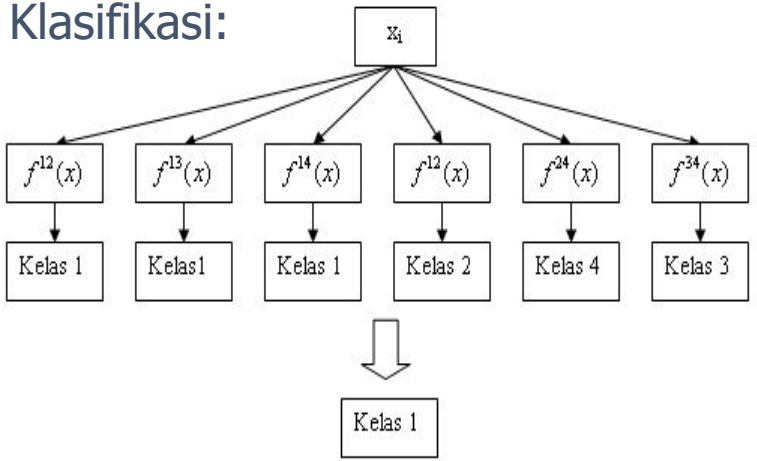
Multi Class SVM: One-against-one

- Dibangun $\frac{k(k-1)}{2}$ model klasifikasi (k adalah jumlah kelas)
- Setiap pelatihan model menggunakan data dari dua kelas
- Prediksi Kelas data dengan metode voting

Pembelajaran:

$y_i = 1$	$y_i = -1$	Hipotesis
Kelas 1	Kelas 2	$f^{12}(x) = (w^{12})x + b^{12}$
Kelas 1	Kelas 3	$f^{13}(x) = (w^{13})x + b^{13}$
Kelas 1	Kelas 4	$f^{14}(x) = (w^{14})x + b^{14}$
Kelas 2	Kelas 3	$f^{23}(x) = (w^{23})x + b^{23}$
Kelas 2	Kelas 4	$f^{24}(x) = (w^{24})x + b^{24}$
Kelas 3	Kelas 4	$f^{34}(x) = (w^{34})x + b^{34}$

Klasifikasi:



Multi Class SVM: DAG (Directed Acyclic Graph)

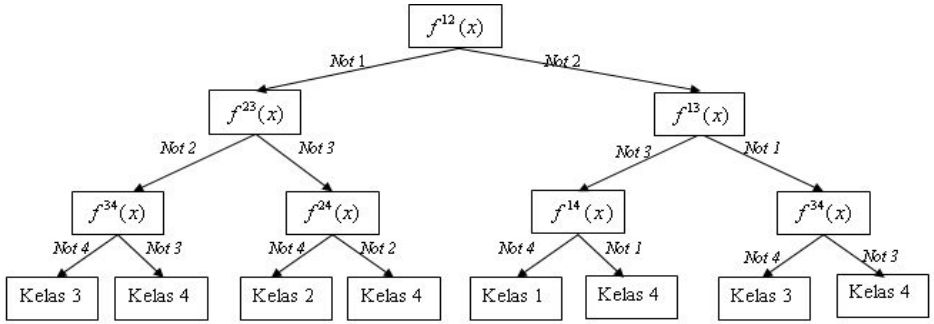
- Proses pembelajaran sama dengan One-Against-One

Pembelajaran:

f12 :
y=1 : bukan kelas 2 (tapi kelas 1)
y=2 : bukan kelas 1 (tapi kelas 2)

$y_i = 1$	$y_i = -1$	Hipotesis
Bukan Kelas 2	Bukan Kelas 1	$f^{12}(x) = (w^{12})x + b^{12}$
Bukan Kelas 3	Bukan Kelas 1	$f^{13}(x) = (w^{13})x + b^{13}$
Bukan Kelas 4	Bukan Kelas 1	$f^{14}(x) = (w^{14})x + b^{14}$
Bukan Kelas 3	Bukan Kelas 2	$f^{23}(x) = (w^{23})x + b^{23}$
Bukan Kelas 4	Bukan Kelas 2	$f^{24}(x) = (w^{24})x + b^{24}$
Bukan Kelas 4	Bukan Kelas 3	$f^{34}(x) = (w^{34})x + b^{34}$

Klasifikasi:



Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Latihan

1. Suatu persoalan klasifikasi 5 kelas akan diselesaikan dengan Multi-class SVM. Berapa model biner pada one-against-all dan one-against-one?

Institut Teknologi Bandung www.itb.ac.id @itb1920 @itbofficial

Thank You

